Multi-UAV Source Seeking and Optimization

BTP Project

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Continuing BTP-1 work....Recap

What was done in BTP-1:

- Last sem the work focused on understanding different PSOs.
- Implementing [2].
- Doing exploring based source seeking for APSO, ARPSO, SPSO.

What is plan for BTP-2:

- This sem target is to try exploitation strategies in source seeking algos and observe behaviours of different PSOs
- To identify which PSO or combination of different PSOs work best in which scenario.

PSO

PSO is the technique used to find an approximate solution for maximization or minimization problems.

- PSO is initialized with a group of random particles (solutions) and then searches for the optimal solution by updating generations.
- Particles move through the solution space, and are evaluated according to some fitness criterion after each time step. In every iteration, each particle is updated by following two "best values".
- The first one is the best solution (fitness) it has achieved so far (the fitness value is also sorted). This value is called P_{best}.
- Another best value that is tracked by the PSO is the best value obtained so far by any particle in the population. This second-best value is a **global best** and called g_{best}.

Objective

Source Seeking:

- Understand [2] and reproduce its results.

Optimization:

- Using CEC2022 benchmarks [1] for optimization of [2] PSO algorithms.
- To develop and evaluate hybrid Particle Swarm Optimization approaches that combine the strengths of different PSO variants
- To improve optimization performance on complex benchmark functions through strategic algorithm hybridization
- To analyze how different hybridization strategies affect convergence behavior and solution quality
- To identify which hybrid approaches are most effective for specific types of optimization problems

APSO (Acceleration based PSO)

- Uses acceleration as the primary update mechanism (second-order dynamics)
- Maintains momentum through previous movement history
- Produces smoother trajectories with more consistent direction

Mathematical Formulation:

- **acc update:** $a_i(t+1) = \omega_1 a_i(t) + c_1 r_1 (p_i - x_i) + c_2 r_2 (g - x_i)$
- **Velocity update:** $v_i(t+1) = \omega_2 v_i(t) + a_i(t+1)T$
- **Position update:** $x_i(t+1) = x_i(t) + v_i(t+1)T$

SPSO (Standard PSO)

- Direct velocity-based update mechanism (first-order dynamics)
- Balanced exploration-exploitation trade-off
- Widely used baseline algorithm in swarm intelligence

Mathematical Formulation:

- **Velocity update:** $v_i(t+1) = \omega_1 v_i(t) + c_1 r_1 (p_i - x_i) + c_2 r_2 (g - x_i)$
- **Position update:** $x_i(t+1) = x_i(t) + v_i(t+1)$

ARPSO (Adaptive Random PSO)

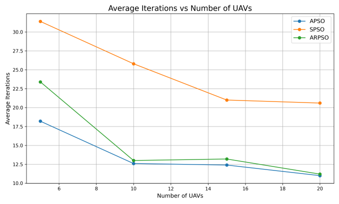
- Incorporates adaptive parameters and randomness
- Dynamically adjusts inertia weight based on swarm state
- Includes additional random attractive positions for exploration

Mathematical Formulation:

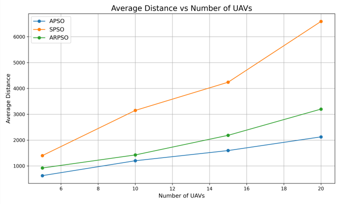
- **Adaptive inertia:** $\omega_i(t+1) = f(evolutionaryspeed, aggregationdegree)$
- **Velocity update:** $v_i(t+1) = \omega_i v_i(t) + c_1 r_1(p_i - x_i) + c_2 r_2(g - x_i) + c_3 r_3(a_i - x_i)$
- here a_i is a random attractive position
- **Position update:** $x_i(t+1) = x_i(t) + v_i(t+1)$

NOTE: For code all the values were taken from the paper [2].

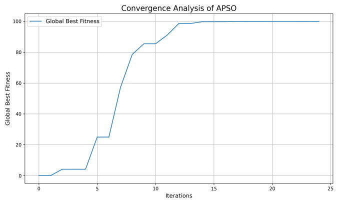
Source Seeking without noise



(a) iterations vs No. of UAV's



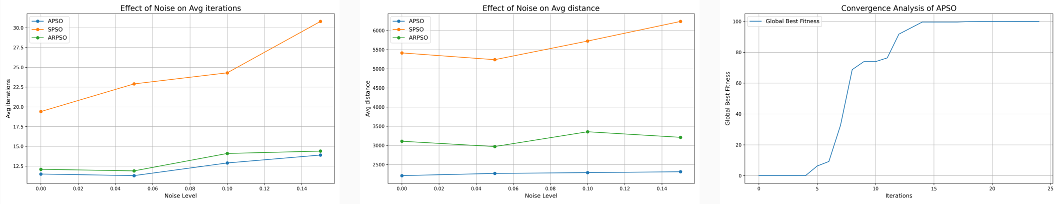
(b) Distance vs No. of UAV's



(c) Convergence analysis

Figure 1: Source Seeking without Noise

Source Seeking with noise



(a) iterations vs Noise

(b) Distance vs Noise

(c) Convergence analysis

Figure 2: Source Seeking with Noise for 30 UAV's

Source Seeking Conclusions

- APSO consistently outperforms both SPSO and ARPSO across all noise levels.
  - APSO requires fewer iterations
  - APSO travels less distance
- APSO shows the best scaling as UAV count increases, iterations actually decrease and SPSO shows the worst scaling in terms of distance traveled.
- All algorithms show degraded performance as noise increases
  - The number of iterations increases with noise level
  - The distance traveled also generally increases with noise level
- APSO shows the best robustness to noise
  - The slope of the APSO line is less steep than SPSO, indicating better resilience to noise
  - Even at 0.15 noise level, APSO maintains good performance
- APSO provides smoother convergence under noise. As its acceleration based approach allows it to maintain momentum through noisy measurements, while the standard PSO gets easily misled by noise.

CEC'22 Benchmark Suit

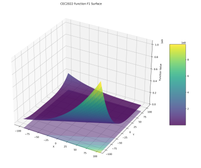
What are CEC Benchmarks?

- Standard test suite developed by IEEE Congress on Evolutionary Computation
- CEC 2022 includes diverse optimization problems with known properties
- Designed to rigorously evaluate and compare optimization algorithms

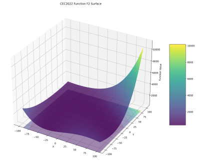
Functions Implemented:

- **F1 (Zakharov):** Continuous unimodal function with interdependent variables
- **F2 (Rosenbrock):** Continuous function with narrow curved valley
- **F3 (Schaffer):** Highly multimodal function with many local optima
- **F4 (Step Rastrigin):** Discontinuous multimodal function with plateaus
- **F5 (Levy):** Multimodal function with many local minima
- **all** 12 functions were implemented from [1]

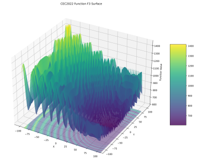
CEC'22 Visualization



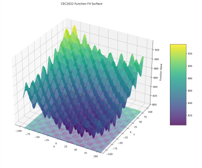
F1



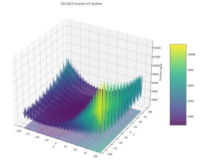
F2



F3



F4



F5

Function surfaces for CEC benchmark functions

CEC2022 Benchmark

Overview:

- The performance of the APSO was compared against the SPSO on the CEC 2022 benchmark suite.
- To ensure a fair comparison, a Bayesian Optimization hyperparameter tuning process was conducted to find a robust set of parameters for each algorithm.
- The tuned algorithms were then benchmarked across 12 functions in 2, 10, and 20 dimensions.
- The results are summarized by ranking the algorithms based on their mean final fitness over 20 independent runs, where Rank 1 indicates superior performance.

* *Performance:* refers to Effectiveness of an PSO. (i.e # of iteration is fixed, the algos that achieves the closest minima for a given function in those iteration is considered better)

Hyperparameter Tuning: Bayesian Optimization i

How it Works:

- **Intelligent Search:** It's a smart, sequential process that decides the next best set of hyperparameters to test based on the results of previous tests.
- **Surrogate Model:** It builds a probabilistic "performance map" (a Gaussian Process) to predict how different params combinations will perform.
- **Acquisition Function:** It uses this map to balance *exploitation* (testing near the current best-known params) and *exploration* (testing in regions with high uncertainty).

Hyperparameter Tuning: Bayesian Optimization ii

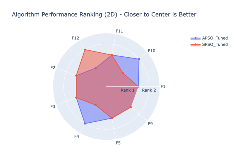
Advantage: It focuses the search on the most promising regions of the vast hyperparams space, leading to a more robust and well-generalized set of optimal params.

Final Tuned Parameters:

- **APSO:** `w1=0.8, w2=-0.285, c1=2.0 c2=1.193`
- **SPSO:** `w=0.6, c1=1.5, c2=1.5`

CEC2022 Observations: 2D i

Performance in Low-Dimensional Space (2D):



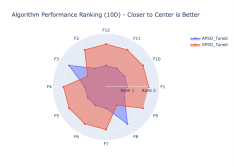
- In 2D tests, neither algorithm demonstrated consistent superiority.
- SPSO achieved Rank 1 on functions F4 and F10, while APSO achieved Rank 1 on function F12.

CEC2022 Observations: 2D ii

- For a majority of the fns, the 2 algos performed almost identically, resulting in a tied rank.
- *Conclusion:* In low-dim search spaces, the additional complexity of APSO's 2nd order dynamics doesn't provide a significant advantage over the simpler, more direct search strategy of SPSO. Both algorithms are effective at finding solutions.

CEC2022 Observations: 10D i

Performance in Moderate-Dimensional Space (10D):

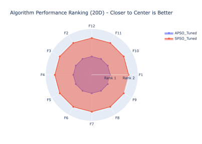


- Distinct performance gap emerged in 10D tests. The APSO algorithm is winner.
- APSO achieved Rank 1 on 10 out of the 12 benchmark functions.

CEC2022 Observations: 10D ii

- *Conclusion:* As the complexity of the search space increases, the theoretical advantages of APSO begin to emerge. Its acceleration based updates appear to enable more effective navigation of the multi-modal and complex landscapes present in 10D, allowing it to find superior solutions compared to SPSO.

Performance in High-Dimensional Space (20D):



- In the 20D tests, the performance gap widened into a complete dominance by APSO.
- APSO achieved a perfect result, securing Rank 1 on all 12 benchmark functions.

CEC2022 Observations: 20D ii

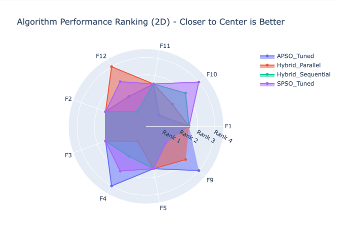
- *Conclusion:* In highly complex, high dim search spaces, the APSO is superior to SPSO. This result strongly suggests that the smoother, momentum based trajectories enabled by 2nd order dynamics are critical for escaping the local optima and effectively exploring the vast search space characteristic of 20D problems. The direct velocity updates of SPSO appear insufficient for this task, leading to premature convergence on suboptimal solutions.

Hybrid Exploitation Strategies

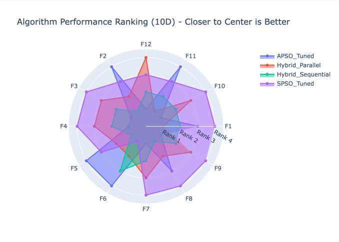
To investigate balance bw global exploration and local exploitation, 2 hybrid strategies were tested.

- **Hybrid Parallel PSO**
  - **Concept:** A single swarm with a *division of labor*.
  - **Implementation:** The swarm of 30 particles was divided into two sub-groups:
    - 15 "Explorer" particles using the **APSO** update rule.
    - 15 "Exploiter" particles using the **SPSO** update rule.
  - **Synergy:** Both sub-groups operate within the same swarm, sharing a single global best solution (g_{best}). This allows explorers to find promising new regions and the exploiters to rapidly converge on them.

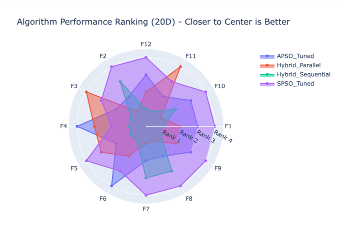
- **Hybrid Sequential PSO**
  - **Concept:** A phased approach, prioritizing exploration before exploitation.
  - **Implementation:** The entire swarm of 30 particles operates in two distinct phases:
    - **Phase 1 (Exploration):** For the first half of the iterations (e.g., 1-250), all particles use the *APSO* update rule to broadly search the landscape.
    - **Phase 2 (Exploitation):** For the second half (e.g., 251-500), all particles switch to the *SPSO* update rule to aggressively fine-tune the best solution found in Phase 1.



2Dim



10Dim



20Dim

Radar Charts

Exploitation Conclusions/Observations

- **Analysis of 2D Results**
  - **Observations:** The performance is very competitive and messy. No single algorithm dominates. Hybrid Sequential gets the most Rank 1s (2), but SPSO Tuned, APSO Tuned, and Hybrid Parallel also get one Rank 1 each. All algorithms are frequently clustered in the middle ranks (2-3).
  - **Conclusion:**
    - **Low-Dimensionality is a "Solved Problem" for PSO.** In a simple 2D search space, all variants of PSO are reasonably effective. The differences between them are marginal. The problem is not complex enough to clearly separate the performance of a good algorithm from a great one.
    - **Hybrid Sequential Shows Promise.** Even in this simple space, the strategy of exploring first (APSO) and then exploiting (SPSO) seems to give a slight edge, securing wins on F12 and F9. This suggests the strategy is fundamentally sound.

- **Analysis of 10D Results**
  - **Observations:** The results are clearer here.
    - APSO is the standout winner, achieving Rank 1 on 6 different functions.
    - Hybrid Sequential is the consistent runner-up, achieving Rank 2 on 8 functions and winning on 3.
    - Hybrid Parallel performs well, securing 2 wins.
    - SPSO performs very poorly, frequently coming in last (Rank 4).
  - **Conclusion:**
    - **APSO's Dynamics Excel in Moderately Complex Spaces.** This result vindicates the theory behind APSO. When tuned properly, its smoother, acceleration-based movement is superior for navigating the complex landscape of a 10D problem compared to the more aggressive SPSO.
    - **Simple Hybridization Isn't a Silver Bullet.** The Hybrid Parallel model, while good, is often beaten by the pure APSO Tuned. This suggests that simply having a mix of particles isn't always better than a single, well-suited strategy. The Hybrid Sequential model's consistency as a runner-up shows its strength, but it can't beat the pure APSO at its own game in this dimension.

- **Analysis of 20D Results**
  - **Observations:** The results are completely different from the 10D case.
    - Hybrid Sequential is the dominant, runaway winner. It achieves Rank 1 on an incredible 8 out of 12 functions. This is a stunning victory.
    - Hybrid Parallel is the clear second-best algorithm, securing the remaining 4 Rank 1s and consistently achieving Rank 2.
    - APSO Tuned, the winner in 10D, now performs poorly, often ranking 3rd or 4th.
    - SPSO Tuned is the worst-performing algorithm, consistently ranking last.
  - **Conclusion:** A Phased Approach is Crucial for High-Dimensionality. In the immense, hyper-complex 20D search space, a single strategy is not enough.
    - Pure APSO gets lost; its global exploration is good, but it fails to effectively fine-tune solutions.
    - Pure SPSO is too aggressive; it gets trapped in local optima immediately and never finds the global basin of attraction.
    - The Hybrid Sequential strategy is perfect. It uses APSO's strength for the first half of the search to navigate the vast space and locate the promising region of the global optimum. Then, it switches to SPSO's strength to aggressively and rapidly converge on the exact solution. This combination of global exploration followed by local exploitation is the winning formula.

Findings

- The Hybrid Sequential strategy is the most robust and powerful algorithm overall, demonstrating overwhelming superiority in the most challenging 20D problems. Its phased approach of exploration then exploitation is the key to its success.
- The effectiveness of a PSO variant is highly dependent on the problem's dimensionality. The best algorithm for 10D (APSO Tuned) was not the best for 20D (Hybrid Sequential), highlighting the **No Free Lunch Theorem.**
- Simple hybridization is not a guaranteed improvement, but structured hybridization is. The Hybrid Parallel model was good, but the more structured Hybrid Sequential model was exceptional, proving that how you combine algorithms matters more than just the fact that you are combining them.
- Proper hyperparameter tuning is essential. The tuned versions of APSO and SPSO dramatically changed the performance landscape compared to the paper's default parameters, validating the entire tuning process you undertook.

Codebase

- github.com/pingu-73/vigilant-spork
- fix-branch: for CEC and exploitation

References

- Wenjian Luo, Xin Lin, Changhe Li, Shengxiang Yang, and Yuhui Shi.
  Benchmark functions for cec 2022 competition on seeking multiple optima in dynamic environments, 2022.
- Adithya Shankar, Himanshu, Harikumar Kandath, and J. Senthilnath.
  Acceleration-based pso for multi-uav source-seeking.
  In *IECON 2023- 49th Annual Conference of the IEEE Industrial Electronics Society*, pages 1–6, 2023.